National Comprehensive Cancer Network®

2026

NCCN Guidelines for Patients®

Cancer care recommendations from leading experts at the National Comprehensive Cancer Network® (NCCN®)

Metastatic Non-Small Cell Lung Cancer

Presented with support from NATIONAL COMPREHENSIVE CANCER NETWORK® FOUNDATION Guiding Treatment. Changing Lives.

Available online at NCCN.org/patientguidelines

NCCN Guidelines for Patients®

The essential guide for people facing cancer.

Based on care recommendations from leading cancer experts.

Explains high-quality cancer care provided at state-of-the-art cancer centers.

Reviewed and revised every year.

Did you know that top cancer centers across the United States work together to improve cancer care? This alliance of leading cancer centers is called the National Comprehensive Cancer Network® (NCCN®).

Because cancer care is always evolving, NCCN develops and frequently updates evidence-based cancer care recommendations used by health care providers worldwide. These recommendations are known as the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®).

The NCCN Guidelines for Patients plainly explain these expert recommendations, so you can talk with your care team about the best care for you.

These NCCN Guidelines for Patients are based on the NCCN Guidelines® for Non-Small Cell Lung Cancer, Version 5.2026 — March 13, 2026.

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National Comprehensive Cancer Network (NCCN) and NCCN Foundation
3025 Chemical Road, Suite 100, Plymouth Meeting, PA 19462 USA

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About metastatic NSCLC

5 What is metastatic NSCLC?

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7 How can I get the best care?

Non-small cell lung cancer (NSCLC) is the most common type of lung cancer. Treatment for widespread NSCLC combined with supportive care improves lives. This guide explains how the growth of widespread cancer can be slowed and how symptoms can be eased.

What is metastatic NSCLC?

Metastatic non-small cell lung cancer (NSCLC) is an abnormal spreading of certain types of lung cancer cells into the lining of the lung or to another organ.

This description is important because treatment differs between lung cancer types and where lung cancer is in the body.

Types of NSCLC

NSCLC is a group of cancers. Each cancer in this group affects a different type of lung cell.

You have NSCLC if you have one of the following lung cancers:

- Adenocarcinoma (A-deh-noh-KAR-sih-NOH-muh) affects cells in the airways that make mucus.
- Large cell carcinoma (kar-sin-OH-ma) forms from large cells in the airways.
- Squamous (SQUAY-mous) cell carcinoma affects cells that form on airways in response to toxic exposure.

There are more types of NSCLC, but they are much less common. Some lung cancers are a mix of cell types.

Metastatic sites of NSCLC

Lung cancer is a disease that causes lung cells to grow out of control and form a tumor on the wall of the airway.

Airways of the lungs

The airways are tube-shaped passages that branch out like a tree in your lungs. Non-small cell lung cancer causes cells from the inner wall of the airways to grow out of control.

Lung cancer cells can also break away from a lung tumor, spread outside of the lung, and form more tumors. Cancer that has spread from where it started is called metastatic.

NSCLC can spread to many different places. It often spreads to the brain, liver, bone, and adrenal glands. NSCLC also spreads into the tissue lining around the lung and from one lung to the other.

How is metastatic NSCLC treated?

Treatment of metastatic NSCLC is based on features of the cancer, your overall health, and your goals for treatment. There's no single treatment plan that's best for everyone.

Testing is needed to plan treatment

Your care team will need to know the type of lung cancer and where the cancer is. They will also test for important cancer features called biomarkers. To see a list of tests needed to plan treatment, read *Chapter 2: Testing for metastatic NSCLC*.

Supportive care addresses the challenges of cancer

Supportive care involves multiple types of aid that improve quality of life. Receiving supportive care early has been shown to extend and enhance life for people with lung cancer. Read more about this important part of cancer care in *Chapter 3: Improving life with supportive care*.

Confused about lung cancer?

People use the term lung cancer to describe many cancers. Sometimes, they misuse the term, so here's an explanation of what lung cancer is and what it isn't.

What lung cancer is

Simply, lung cancer is a disease that causes uncontrolled growth of lung cells. There are many types of lung cells, so there are many types of lung cancer.

Non-small cell lung cancer (NSCLC) is the most common type of lung cancer. So when people talk about lung cancer, they're likely talking about NSCLC.

What lung cancer isn't

Cancers that have spread to the lungs are not lung cancers. For example, stomach cancer that has spread to the lungs is still stomach cancer.

In addition to this guide, the library of NCCN Guidelines for Patients has guides for metastatic NSCLC, other types of lung cancer, and cancers that have spread to the lungs. The library can be found at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.

Systemic therapy is the most common treatment

Systemic therapy consists of prescribed drugs that treat cancer throughout the entire body. It's the focus of this guide because it's used to treat most metastatic lung cancers.

Less often, local treatment is used to try to cure limited metastases. For example, surgery or radiation therapy may be done to control lung cancer that has spread only to an adrenal gland.

People with metastatic lung cancer are often on systemic therapy for the rest of their lives. Typically, a person takes one treatment then switches to another if the cancer grows again. The types and order of systemic therapy are discussed in Chapters 4 through 6. Each chapter focuses on a specific biomarker type of NSCLC.

Clinical trials offer hope

Clinical trials are a type of health research that aim to find better ways of treating cancer. Often, standard treatment is compared to a promising new drug. A clinical trial may be an option for you. Learn more about clinical trials in *Chapter 6: Treatment by NSCLC type*.

How this guide can help you

Making decisions about cancer care is stressful. There's a lot to learn, and you don't know what the future holds.

Use this guide to get the information and support you need.

Patients, doctors, and other health care professionals trust the NCCN Guidelines for Patients. This guide uses clear, everyday language to explain current cancer care recommendations made by respected experts in the field. Their recommendations are based on the latest research and practices at leading cancer centers.

Your health is unique to you, so your cancer care should be, too. As you read this guide, you'll learn which treatments are likely to provide the best results for you. And you'll be better prepared to talk with your care team

How can I get the best care?

Advocate for yourself. You have an important role to play in your care. Many people feel more satisfied when they actively take part in planning their cancer care.

The NCCN Guidelines for Patients will help you play a larger role in your care. Discuss the recommendations in this guide with your care team. Ask questions about your options and share your goals and concerns.

Don't know what to ask? You're not alone. That's why we include suggested questions to ask at the end of chapters.

Keep reading to find the best care for you.

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Several tests are needed to confirm lung cancer and plan treatment. Your care team will test for the type of lung cancer, extent of the cancer, and for cancer markers. They will also collect information on your overall health.

Many times, metastatic non-small cell lung cancer (NSCLC) is suspected because of concerning symptoms. Less often, a mass that may be cancer is found by accident on a scan for another health issue. In this chapter, you'll learn how your care team:

- Decides where and how to remove tissue samples for cancer testing
- Assesses the extent, or stage, of the cancer in your body
- Learns the type and features of the cancer
- Evaluates your overall health

It takes a team

Lung cancer is confirmed, or diagnosed, by testing samples of body tissue or fluid for cancer cells. How tissue is removed for testing differs between people. It partly depends on where the cancer might be in your body.

Lung cancer stages and metastatic cancer

There are 4 main stages of lung cancer. You may see them written with Roman numerals—stages I (1), II (2), III (3), and IV (4). Higher stages mean a greater extent of cancer.

Stage 4 is metastatic cancer at diagnosis. The cancer has spread to the lining of the lung or to other organs. Sometimes, earlier stages of lung cancer become metastatic after treatment.

A team of experts will decide the best way to confirm if you have lung cancer. Your diagnostic team may include these experts:

- A pulmonologist is an expert in lung diseases.
- A thoracic radiologist is an expert in reading pictures (images) of the inside of the chest.
- A thoracic surgeon is an expert in performing operations within the chest.
- An interventional radiologist is an expert in performing image-guided biopsies.
- A pathologist is an expert in tissue and cells and confirming cancer.

You are an important part of the team. Your input is just as important as tests for planning treatment.

Detecting cancer on imaging

Imaging shows the inside of the body and can show where the cancer may be. Your team will use imaging to decide the best place to remove tissue samples for cancer testing.

Your team will also use imaging for cancer staging. A cancer stage describes the extent of lung cancer in the body. For many people, diagnosis and cancer staging are done at the same time. You will likely have 3 types of imaging.

CT scan

Computed tomography (CT) is often the first scan done to stage lung cancer. Your team will look for tumors in images of your chest and upper abdomen, including the adrenal glands.

A CT scan is a more detailed kind of x-ray. It takes many pictures from different angles. A computer combines the images to make 3D pictures.

You'll receive an injection of contrast if it's safe for you. Contrast is a substance that makes images clearer.

FDG-PET/CT scan

Positron emission tomography (PET) combined with a CT scan is necessary if it has not been done before. PET highlights tissue in your body that may be cancerous. A PET/CT scan may detect cancer that was not found by CT alone or may show signs that the cancer has spread.

Your whole body may be scanned, or the scan may extend from your neck to the middle of your thighs.

Before the scan, you will be injected with a tracer called fluorodeoxyglucose, or FDG. The tracer makes cancer cells show up as bright (or hot) spots on the scan. Multiple health problems can cause hot spots, so hot spots that suggest cancer often need to be confirmed by other tests.

Imaging

Imaging is used to make pictures of the inside of your body. During the scan, you will lie on a table that will move into the tunnel of the machine. After the scan, a radiologist will view the pictures on a computer and look for signs of cancer.

Brain MRI

Lung cancer tends to spread to the brain, so it's very important to test if it has. Magnetic resonance imaging (MRI) may show small brain tumors that aren't causing symptoms.

MRI uses a safe magnetic field and radio waves to make pictures. You'll have a scan with contrast and another scan without contrast unless contrast is not safe for you. If you can't have an MRI, you may get a CT scan of your head with contrast.

"The good news is that today the medical industry has made great advances in treating cancer. They create a custom designed treatment specifically for you."

Confirming cancer with biopsy

A biopsy is a procedure that removes samples of body tissue or fluid for cancer testing. Often, tissue from the organ with the metastasis is taken rather than from the lungs. Your team will use imaging to select the biopsy site, which is often the adrenal gland, liver, or bone.

Types of biopsy

The type of biopsy that will be done depends on the body part and the experience of your team. Common types of biopsies for metastatic lung cancer are:

- **An external needle biopsy** involves guiding a thin needle through your skin and into a tumor. These biopsies include transthoracic needle aspiration, core needle biopsies, pericardiocentesis, and thoracentesis.
- **Down-the-throat biopsies** is when a thin tube is guided down your throat into your airway (bronchus) or food pipe (esophagus). These procedures include many types of bronchoscopy.
- **Keyhole surgeries** involve making small openings in your chest. Small tools are inserted through the holes to remove tissue. This method is less invasive than open surgery. These surgeries include laparoscopy and thoracoscopy. Thoracoscopy is also called video-assisted thoracoscopic surgery.

Pathology review of biopsy samples

A pathologist will prepare and then look at the tissue with a microscope. This takes a few days. The remaining tissue will be saved for possible future testing.

If NSCLC is found, the pathologist will identify the type.

- Adenocarcinoma is a cancer of cells that often line the air sacs in the lungs and make mucus. This is the most common type of NSCLC.
- Large cell carcinoma is a cancer of cells that are larger than other cells in the airways.
- Squamous cell carcinoma is a cancer of thin, flat cells and often occurs in the central part of a lung.
- There are also rare and mixed types of NSCLC.

The results of lab tests used for diagnosis are recorded in a pathology report.

Planning treatment

After metastatic NSCLC is confirmed, your care team will focus on treating the cancer. One member of your team will be a medical oncologist. A medical oncologist is trained in treating cancer with prescribed drugs.

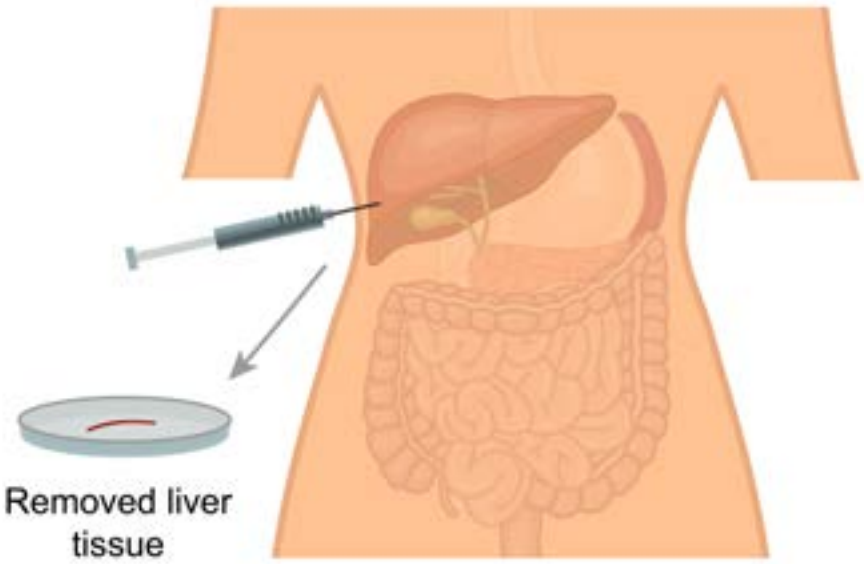
A radiation oncologist may also be part of your team. This specialist treats cancer or relieves symptoms with radiation. Learn more about symptom relief and other team members in *Chapter 3: Improving life with supportive care*.

To plan treatment, your team will need to learn more about the cancer and your overall health. They will use your diagnostic test results and the tests described next in this chapter. A complete list of tests for metastatic NSCLC is in **Guide 1**.

You might need to get new scans. Scans done more than 60 days ago should not be used to decide your treatment.

Biopsy of metastasis

If your care team suspects metastatic lung cancer, you may have a biopsy of the metastasis instead of the tumor in the lung. A biopsy of the metastasis can diagnose and stage the cancer at the same time. A needle biopsy through the skin may reach some metastases, such as in the liver (shown). Imaging is often used to help guide the needle to the correct spot.



Assessing your overall health

Your team will perform a thorough evaluation of your health. They need to know about all of your health issues, both those caused by the cancer and those unrelated to the cancer. They also need to determine all of the places where the lung cancer has likely spread.

Medical history

Expect your team to ask a lot of questions about your health. This is known as taking a medical history. Your team will want to know about your past and current health, such as:

- Illnesses and injuries
- Symptoms like unexplained weight loss, trouble breathing, chest pain, and cough
- Prescribed and over-the-counter medicines and supplements
- Previous surgeries
- Lifestyle choices, including your diet, how active you are, and whether you smoke or drink alcohol

Also be prepared to discuss the health of your close blood (first-degree) relatives. First-degree relatives include siblings, parents, and grandparents related to you by birth.

Physical exam

A team member will also perform a thorough physical exam of your body.

Guide 1
Tests for metastatic non-small cell lung cancer

Imaging	• Diagnostic CT scan of the chest and upper abdomen with contrast • FDG-PET/CT scan • Brain MRI
Cancer tests	• Biopsy of metastasis • Pathology review
Overall health tests	• Medical history, including weight loss and smoking history • Physical exam • Performance status • Complete blood count and chemistry profile • Pulmonary function tests
Biomarker tests	• Molecular tests for driver mutations • Immunohistochemistry for PD-L1 levels • If there's enough tissue for testing, immunohistochemistry for HER2 and HGF receptor (c-Met) overexpression

A physical exam may include:

- Checking your vital signs—blood pressure, heart rate, breathing rate, and body temperature
- Listening to or feeling organs, including your lungs, spleen, and liver
- Feeling for enlarged lymph nodes, which are small disease-fighting structures located throughout your body
- Assessing your level of pain, if any, when you are touched

Performance status

Performance status is your ability to do day-to-day activities. It's based on your medical history and exam. Your team will use your performance status to decide if your body can endure intense treatment.

Blood tests

Blood tests are commonly used to screen for disease. They are also used to assess if cancer is affecting organs. Samples of your blood will be needed for these blood tests:

- A **complete blood count** measures parts of the blood, including counts of white blood cells, red blood cells, and platelets.
- A **chemistry profile** assesses sugar and natural salts in your body and how well your liver and kidneys are working.

Pulmonary function tests

Pulmonary function tests are breathing activities used to assess how well your lungs work. Your team will use the results to decide if certain treatments are safe.

Profiling cancer with biomarker tests

Biomarker tests look for biological clues, or markers, that are used to make decisions about cancer care. Biomarkers differ between types of cancer and among people with the same type of cancer.

Many biomarkers of lung cancer help the cancer grow. There are treatments for some biomarkers, such as those listed in **Guide 2**.

Biomarker tests are performed on tumor tissue removed during a biopsy or surgery. A blood sample also may be tested.

Driver mutations

One type of biomarker is a driver mutation. Driver mutations are abnormal genes in DNA that boost cancer growth. These mutations are detected with molecular tests.

Biomarker testing for driver mutations is recommended for adenocarcinoma, large cell carcinoma, and rare cell types. The decision to test squamous cell lung cancer for driver mutations is decided on a person-by-person basis.

NCCN experts strongly advise the use of multigene panel testing when testing for driver mutations. This testing looks for many genes at the same time, so it can take up to 3 weeks to get the results.

Guide 2
Biomarkers of metastatic NSCLC

Driver mutations

- *EGFR* exon 19 deletion or L858R mutation
- *EGFR* S768I, L861Q, or G719X mutation
- *EGFR* exon 20 insertion
- *KRAS* G12C mutation
- *ALK* gene fusion
- *ROS1* gene fusion
- *BRAF* V600E mutation
- *NTRK* gene fusion
- *MET* exon 14 skipping
- *RET* gene fusion
- *HER2* (*ERBB2*) mutation
- *NRG1* gene fusion

Protein expression

- PD-L1 levels
- HER2 overexpression
- HGF receptor (c-Met) overexpression

The panel should test at least for:

- *EGFR* exon 19 deletion and *MET* exon 14 skipping—both of these driver mutations have a missing part of the gene
- *EGFR* L858R, S768I, L861Q, and G719X, *KRAS* G12C, BRAF V600E, HER2 (ERBB2) mutation—these show that a change has occurred in the gene's normal DNA
- *EGFR* exon 20 insertion—a new part has been added to the gene
- *ALK*, *ROS1*, *NTRK*, *RET*, and *NRG1* gene fusions—a piece of these genes has joined to a piece of another gene, creating an abnormal new gene

Protein expression

Another type of biomarker are proteins on the surface of the cancer cells. The level of these proteins, called expression, is detected with a lab test called immunohistochemistry. High levels are referred to as overexpression.

- All lung cancers should be tested for PD-L1 levels. (For more information on PD-L1, read Chapter 5.)
- Lung adenocarcinoma, large cell carcinoma, and rare cell types may be tested for HER2 and HGF receptor (c-Met) overexpression before treatment starts if there's enough tissue.
- Testing for HER2 overexpression in squamous cell lung cancer is decided on a person-by-person basis.

Key points

- Imaging is used to show where lung cancer may be in your body.
- To diagnose and stage metastatic lung cancer, a body part that appears to have cancer and is far from the lung tumor will likely be sampled and tested.
- To plan treatment, your care team will ask about your health, examine your body, and test blood samples.
- Biomarker tests look for small yet important features of lung cancer that differ among people. There are treatments for some markers.

Questions to ask

- Could the tests be wrong or cause health problems?
- Will I need to pay for any of the tests?
- What do I need to do to prepare for testing?
- What are the procedures for removing and storing tissue samples for future testing?
- How can I get a copy of the test results?

What's next?

Your team will explain the test results to you and what they recommend for your care. A care plan usually includes treatment for cancer and support for you. The next chapter is about supportive care. It's important to start supportive care early.

Let us know what you think!

Please take a moment to complete an online survey about the NCCN Guidelines for Patients.
NCCN.org/patients/response

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Improving life with supportive care

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Metastatic lung cancer often has a major impact on the body. Supportive care, especially when started early, can improve your quality of life. A key part of care is reducing symptoms.

The main concern for most people with cancer is to get treatment that works. But having cancer often involves additional challenges to one's health and life. This chapter discusses what help is available for these challenges.

What is supportive care?

Supportive care is an important part of cancer care. The goal is to improve your quality of life during and after cancer treatment. Supportive care is for everyone with cancer and their families, not just for those at the end of life. It's also known as palliative care.

Supportive care includes a wide range of services. Supportive care prevents or manages the symptoms of cancer and the side effects of cancer treatment, like pain and cancer-related fatigue. It also addresses the mental, social, emotional, and spiritual concerns faced by people with cancer.

Supportive care provides help with additional needs, such as:

- Making treatment decisions
- Coordinating your care
- Paying for care
- Planning for advance care and end of life

"A common myth is that palliative care is only for terminally ill patients. It is so much more! It is worth reaching out to palliative care in your hospital or clinic. They treat the whole patient, not just cancer."

Starting supportive care early

NCCN experts recommend combining supportive care with cancer care shortly after diagnosis of non-small cell lung cancer (NSCLC). Many studies have shown that receiving specialized supportive care early can be more helpful than starting supportive care later. Early supportive care can extend and enhance life for people with NSCLC.

Palliative care team

A palliative care specialist may be a member of your care team. This specialist has received specific training to provide maximal supportive care to you. They have a holistic approach to managing lung cancer symptoms and work with the other specialists on your team.

Other specialists who may be involved in your care include a:

- Radiation oncologist
- Respiratory therapist
- Rehabilitation specialist
- Registered dietitian
- Social worker
- Chaplain

Help to quit smoking

Smoking can limit how well cancer treatment works.

- If you don't smoke, it's important that you don't start now.
- If you smoke, it's important to quit. It's never too late.

Nicotine addiction is one of the hardest addictions to stop. The stress of having cancer may make it even harder to quit.

The *NCCN Guidelines for Patients: Quitting Smoking* provides critical support and guidance for people with cancer. This guide explains how best to use the tools that exist to help you quit for good.

If you tried to quit before, try again. Most people slip or relapse before quitting for good.

"It is important to understand what you are going through. If your doctor says something you do not understand, don't just nod like you do. Let your doctor know. You can say, 'I don't fully understand what you just said. Can you please explain it in simpler terms?'"

Lung cancer complications

People with metastatic NSCLC often have health issues, called complications, that are caused by the cancer. Your care team will create a management plan based on your needs.

Local treatment is commonly used to reduce complications caused by metastasis. It's called local treatment because it focuses on a specific area of cancer.

There are 3 types of local treatment. Surgery may be used to remove some lung cancer. Radiation therapy kills an area of cancer with very precise, high-dose beams. Chemoradiation combines chemotherapy and radiation therapy.

Local treatment is used to:

- Improve breathing if lung cancer is blocking the airways
- Improve eating if lung cancer is blocking the digestive tract
- Reduce coughing up blood caused by lung cancer
- Prevent and relieve pain caused by lung cancer pressing against tissue and nerves
- Relieve symptoms caused by lung cancer that has spread to the brain

Managing common cancer effects

The library of NCCN Guidelines for Patients has several guides that focus on supportive care. These guides address physical and emotional symptoms that are commonly caused by many types of cancer or cancer treatments. They are briefly described next so you can find the information you need.

Palliative care

The *NCCN Guidelines for Patients: Palliative Care* describe care for physical and emotional symptoms, such as:

- Diarrhea, constipation, and sleep issues
- Stress and grief

It also provides guidance on talking with your team about cancer treatment and advance care planning.

Distress

Everyone with cancer feels distressed at some point. It's normal to feel worried, sad, helpless, or angry. *NCCN Guidelines for Patients: Distress During Cancer Care* empowers people to get help for distress.

Fatigue

Cancer-related fatigue is not the typical tiredness that follows an active or long day. It's a lack of energy that is distressing, does not improve with normal rest or sleep, and disrupts life. Read *NCCN Guidelines for Patients: Fatigue and Cancer* to learn about physical activity and other methods that reduce cancer-related fatigue.

Nausea and vomiting

Both chemotherapy and radiation therapy can cause nausea and vomiting. Nausea is the feeling that you are going to throw up. Vomiting is forcefully throwing up what's in your stomach. Treatments that prevent and manage both conditions are discussed in *NCCN Guidelines for Patients: Nausea and Vomiting*.

Low blood counts

Chemotherapy often causes a drop in blood cells. Depending on the type of blood cell affected, you may have fatigue, unusual bleeding, and more infections than usual. Recommendations on how to manage low counts are described in *NCCN Guidelines for Patients: Low Blood Counts*.

Immunotherapy side effects

Immune checkpoint inhibitors are used to treat many types of NSCLC. However, this treatment may cause your immune cells to attack your healthy cells. *NCCN Guidelines for Patients: Immunotherapy Side Effects: Immune Checkpoint Inhibitors* explains treatment for many types of side effects.

Blood clots

Some drug treatments for NSCLC increase the risk for blood clots. A thrombus is a type of blood clot that can dangerously block blood flow. *NCCN Guidelines for Patients: Blood Clots and Cancer* explains ways to prevent and treat thrombi.

Supportive care resources

The library of NCCN Guidelines for Patients has a series of guides on supportive care, available at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.

Key points

- Supportive care aims to improve your quality of life, including preventing and relieving symptoms.
- Starting supportive care early has more benefits than starting later.
- Quitting smoking may improve treatment results.

Questions to ask

- When will we discuss a supportive care plan for me?
- What are the best methods to help me quit smoking?
- How will you prevent cancer-related health issues and relieve the symptoms I have?
- Should I tell health care providers I'm getting supportive care? They may incorrectly think I'm getting hospice care.
- Who should I contact if my symptoms get worse?

What's next?

The next several chapters explain NCCN recommendations for cancer treatment. Start reading the chapter that pertains to the type of biomarker, if any, the cancer has.

- If the cancer has a driver mutation, start with *Chapter 4: Treatment of driver mutations*.
- Cancers with PD-L1 and no driver mutations are covered in *Chapter 5: Treatment based on abnormal PD-L1 levels*.
- Some lung cancers do not have a driver mutation and have normal PD-L1 levels. Treatments for these cancers are explained in *Chapter 6: Treatment by NSCLC type*.

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31 *BRAF* V600E mutation

32 *NTRK* gene fusion

32 *MET* exon 14 skipping

32 *RET* gene fusion

33 *HER2* mutation

33 *NRG1* gene fusion

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Driver mutations help many lung cancers grow. Medicine that targets these mutations can help slow cancer growth.

What are driver mutations?

A driver mutation is an abnormal change in the DNA of a cell that supports cancer growth. It enables cancer cells to quickly duplicate, survive, and spread in the body.

At this time, several driver mutations in non-small cell lung cancer (NSCLC) have been found. And researchers are looking for more. Recommendations for driver mutation testing are in *Chapter 2: Testing for metastatic NSCLC*.

Treating mutations is more precise

Chemotherapy destroys any fast-growing cells even if they're not cancer cells. It was once the only treatment for metastatic lung cancer.

Today, newer treatments target the effects of driver mutations and harm fewer normal cells. These newer treatments are called targeted therapy.

Types of targeted therapy

Driver mutations create cell proteins that help cancer cells grow. Targeted therapy is a drug treatment that stops these proteins.

Targeted therapy

Targeted therapy is an important treatment for lung cancer with a driver mutation.

Driver mutations help cancers grow. Several types of driver mutations have been found in lung cancer.

Your care team will plan your treatment based on what type of driver mutation is found.

Kinase inhibitors

Kinases are a type of cell protein that are part of many chemical pathways. Some of them start cell growth.

Kinase inhibitors are drugs that stop the activity of kinases and, in turn, lower the number of new cancer cells being made. They are pills that can be taken at home.

Kinase inhibitors are a common treatment of driver mutations. The drugs listed in this chapter that end in -nib or -sib are kinase inhibitors.

Antibody therapy

Cells have receptors on their surface. Cell receptors receive and send signals like antennas. Antibody therapy stops receptor signals.

- EGFR, MET, and HER2/HER3 antibodies attach to protein receptors on cancer cells and stop signals that tell the cancer cells to grow.
- VEGF antibodies stop the growth of blood vessels that feed tumors. Without blood, cancer cells die.

Most antibody therapies are received through a slow drip (infusion) from a needle into a vein.

Antibody-drug conjugate

An antibody-drug conjugate combines 2 drugs in 1 medicine. The antibody finds and binds to certain cancer cells, and then the other drug attacks the cancer. Antibody-drug conjugates are given by infusion.

NCCN levels of preference

The next sections in this chapter list recommended regimens for each type of driver mutation. A regimen is one or more cancer drugs that are taken at a specific dose, schedule, and length of time. You might need to read several sections if the lung cancer has more than one driver mutation.

All regimens listed in this guide are appropriate. When helpful, NCCN experts also assign a level of preference to their recommendations:

- **Preferred therapies** have the most evidence they may work better and be safer than other therapies.
- **Other recommended therapies** can provide effective results but may have less evidence, more side effects, or may not work quite as well as preferred therapies.
- **Therapies used in certain cases** work best for individuals with specific cancer features or health circumstances.

“Keep seeking out information and read information again as the diagnosis and treatment process continues.”

EGFR deletions and mutations

EGFR is a type of receptor on lung cancer cells. When there's a deletion or a mutation in the gene that makes them, EGFRs can become overactive. Overactive EGFRs cause cancer cells to grow quickly.

EGFR-targeted therapy

NCCN experts recommend targeted therapy to treat NSCLC with *EGFR* deletions and mutations. If you're getting a different type of treatment, you may stop your current treatment early and start targeted therapy. Options for targeted therapy are given in **Guide 3**.

Guide 3
Targeted therapy for metastatic NSCLC with *EGFR* deletions and mutations

***EGFR* exon 19 deletion or *EGFR* L858R mutation**	**Preferred therapies** • Osimertinib • Carboplatin (or Cisplatin), Osimertinib, Pemetrexed for adenocarcinoma, large cell carcinoma, and rare cell types of lung cancer • Lazertinib plus Amivantamab-vmjw **Therapies used in certain cases** • Afatinib • Dacomitinib • Erlotinib • Erlotinib plus Bevacizumab • Erlotinib plus Ramucirumab • Gefitinib • Lazertinib
***EGFR* S768I, L861Q, or G719X mutation**	**Preferred therapies** • Afatinib • Osimertinib **Other recommended therapies** • Dacomitinib • Erlotinib • Gefitinib

EGFR exon 19 deletion or *EGFR* L858R mutation

Osimertinib (Tagrisso) is a preferred therapy for lung cancers with an *EGFR* exon 19 deletion or *EGFR* L858R mutation. It may be combined with chemotherapy to treat lung adenocarcinoma, large cell carcinoma, and rare cell types. Chemotherapy may consist of either carboplatin and pemetrexed (Alimta, Axtle, Pemfexy) or cisplatin and pemetrexed.

Lazertinib (Lazcluze) plus an antibody called amivantamab-vmjw (Rybrevant) is also a preferred therapy. Instead of infusions, amivantamab and hyaluronidase-lpuj (Rybrevant Faspro) injections under the skin may be an option.

Therapies used in certain cases include afatinib (Gilotrif), dacomitinib (Vizimpro), erlotinib (Tarceva), gefitinib (Iressa), and lazertinib. Erlotinib is sometimes combined with a VEGF antibody—bevacizumab (Avastin) or ramucirumab (Cyramza).

EGFR S768I, L861Q, or G719X mutation

Regimens to treat lung cancer with an *EGFR* S768I, L861Q, or G719X mutation consist of one drug. Preferred therapies are afatinib and osimertinib. Other recommended therapies include dacomitinib, erlotinib, and gefitinib.

Options if NSCLC grows again

Within a few years of starting targeted therapy, lung cancer begins to grow again in most people.

To plan the next treatment, you may need to get a biopsy to check for:

- New mutations that stop targeted therapy from working—a T790M mutation is common after taking erlotinib, afatinib, gefitinib, or dacomitinib
- A change in the type of cancer from adenocarcinoma to small cell lung cancer

The next treatment options are listed in **Guide 4**.

Guide 4
Options if metastatic NSCLC grows during EGFR-targeted therapy

Stay on targeted therapy if it has some benefit. Your care team will consider if local treatment would also help.

Switch to a different targeted therapy

- Switch from Afatinib, Dacomitinib, Erlotinib, or Gefitinib to Osimertinib
- Switch from chemotherapy-Osimertinib to Datopotamab deruxtecan-dlnk (Preferred therapy) or to Lazertinib plus Amivantamab (Other recommended therapy)
- Switch from Osimertinib to Carboplatin, Pemetrexed plus Amivantamab (Preferred therapy) or to Lazertinib plus Amivantamab (Used in certain cases)
- Switch from a kinase inhibitor to Afatinib plus Cetuximab

Start treatment listed in Chapter 6

Adding local treatment

If the cancer did not spread to many more places, your team may recommend local treatment and targeted therapy. Local treatment is used to treat cancer in a specific area or organ.

- Radiation therapy uses very precise, high-dose beams to damage cancer cells.
- Surgery removes tumors or organs with cancer.
- Image-guided thermal ablation therapy uses extreme heat or cold to destroy cancer.

Staying on current targeted therapy

Although the cancer may be growing again, targeted therapy could be slowing its growth. You'll likely stay on your current treatment. It depends on how slowly the cancer is growing or if it's only growing in a few places. The cancer may grow faster if targeted therapy is completely stopped.

Switching targeted therapy

Switching to a different targeted therapy may help, especially if there are new mutations.

Osimertinib after erlotinib, afatinib, gefitinib, or dacomitinib may be an option if there's a T790M mutation. Osimertinib may also be an option if the cancer has spread to the brain or the brain's lining whether there's a T790M mutation or not.

If you had osimertinib with chemotherapy, a preferred treatment is datopotamab deruxtecan-dlnk (Datroway). It's an antibody-drug conjugate. Other recommended therapies include lazertinib plus amivantamab. More options for second-line therapy are discussed in *Chapter 6: Treatment by NSCLC type*.

After treatment with only osimertinib, a different type of targeted therapy may treat lung adenocarcinomas, large cell carcinomas, and rare cell types that have an *EGFR* exon 19 deletion or L858R mutation. The preferred therapy is carboplatin and pemetrexed plus amivantamab. Lazertinib plus amivantamab is used in certain cases.

Afatinib plus an antibody called cetuximab (Erbitux) may be an option after an EGFR-kinase inhibitor.

Treatment by type of NSCLC

If targeted therapy is not likely to help, your team may recommend treatment described in *Chapter 6: Treatment by NSCLC type*.

EGFR insertion

An *EGFR* exon 20 insertion makes EGFR overactive, causing the cancer cells to grow quickly. Treatment is based on the type of NSCLC.

The preferred therapy for adenocarcinoma, large cell carcinoma, and rare cell types is chemotherapy (carboplatin, pemetrexed) with the antibody amivantamab-vmjw (Rybrevant, Rybrevant Faspro).

Otherwise, lung cancers with *EGFR* exon 20 insertion are first treated as explained in *Chapter 6: Treatment by NSCLC type*.

If the cancer grows, amivantamab is an option if you haven't had it before. Sunvozertinib (Zegfrovy) is another option. A third option for adenocarcinoma, large cell carcinoma, and rare cell types is an antibody-drug conjugate called datopotamab deruxtecan-dlnk

(Datroway). Additional options for second-line therapies are listed in Chapter 6.

KRAS G12C mutation

A signaling protein inside of lung cells called KRAS can be overactive, causing the cells to grow quickly. A mutation in the *KRAS* G12C gene causes the KRAS protein to be overactive.

Lung cancers with *KRAS* G12C mutation are first treated based on PD-L1 level.

- For treatment of lung cancer with PD-L1, see *Chapter 5: Treatment based on abnormal PD-L1 levels*.
- For treatment of lung cancer without PD-L1, see *Chapter 6: Treatment by NSCLC type*.

If the cancer grows again, your treatment may be switched to one of the following KRAS inhibitors:

- Sotorasib (Lumakras)
- Adagrasib (Krazati)

If the cancer grows during targeted therapy, the next treatment options are second-line therapies listed in Chapter 6.

ALK gene fusion

For some lung cancers, the ALK surface receptor is overactive, causing tumor cells to grow quickly. Overactivity results from parts of 2 genes switching places and forming a new gene, called a gene fusion.

Starting targeted therapy

NCCN experts recommend targeted therapy to treat NSCLC with an *ALK* gene fusion. If you're getting a different type of treatment, you may stop your current treatment early and start targeted therapy.

ALK inhibitors are used to treat cancer with an *ALK* gene fusion. Preferred therapies are alectinib (Alecensa), brigatinib (Alunbrig), ensartinib (Ensacove), and lorlatinib (Lorbrena). Ceritinib (Zykadia) and crizotinib (Xalkori) are used in certain cases. These options are listed in **Guide 5**.

Guide 5
Targeted therapy for metastatic NSCLC with an *ALK* gene fusion

Preferred therapies	• Alectinib • Brigatinib • Ensartinib • Lorlatinib
Therapies used in certain cases	• Ceritinib • Crizotinib

Options if NSCLC grows again

Within a few years of starting targeted therapy, lung cancer starts to grow again in most people. You may need another biopsy to test for new mutations, which might change your treatment options. Treatment options are given in **Guide 6**.

Adding local treatment

If the cancer did not spread to many more places, your team may recommend local treatment and staying on targeted therapy. Local treatment is used to treat cancer in a specific area or organ.

- Radiation therapy uses very precise, high-dose beams to damage cancer cells.
- Surgery removes tumors or organs with cancer.
- Image-guided thermal ablation therapy uses extreme heat or cold to destroy cancer.

Staying on current targeted therapy

Although the cancer may be growing again, targeted therapy could be slowing its growth. For this reason, you may stay on your current treatment. The cancer may grow faster if targeted therapy is completely stopped.

Switching targeted therapy

Switching to a different targeted therapy may help, especially if there are new mutations. Lorlatinib after alectinib, brigatinib, ceritinib, or ensartinib may be an option if there's a mutation, such as an *ALK* G1202R or L1196M mutation. After taking crizotinib, you may switch to alectinib, brigatinib, ceritinib, ensartinib, or lorlatinib.

Treatment by type of NSCLC

If targeted therapy is not likely to help, your team may recommend the treatments listed in *Chapter 6: Treatment by NSCLC type*.

Guide 6
Options if metastatic NSCLC grows during ALK-targeted therapy

Local treatment of limited tumors may be helpful for some people

Stay on first-line therapy if it has some benefit
- Stay on Alectinib, Brigatinib, Ensartinib, Ceritinib, or Lorlatinib if the cancer didn't spread to many more places
- Stay on Crizotinib if the cancer didn't spread to the brain or many more places

Switch to a newer ALK inhibitor
- Switch to Lorlatinib if there's a mutation that stops Alectinib, Brigatinib, or Ceritinib from working
- Switch from Crizotinib to Alectinib, Brigatinib, Ensartinib, Ceritinib, or Lorlatinib

Start treatment listed in Chapter 6

ROS1 gene fusion

A cell surface receptor called ROS1 can be overactive, causing lung cells to grow quickly. The overactivity is caused by parts of 2 genes switching places and creating a new gene called a gene fusion.

Starting targeted therapy

NCCN experts recommend targeted therapy to treat NSCLC with a *ROS1* gene fusion. If you're getting a different type of treatment, you may stop your current treatment early and start targeted therapy.

Kinase inhibitors that target ROS1 are used for treatment. The preferred therapies are:

- Crizotinib (Xalkori)
- Entrectinib (Rozlytrek)
- Repotrectinib (Augtyro)
- Taletrectinib (Ibtrozi)

Entrectinib, repotrectinib, or taletrectinib may be better for treating lung cancer in the brain.

Options when NSCLC grows again

In time, the cancer will grow despite ongoing targeted therapy. If the cancer didn't spread to many more places, your team may recommend local treatment, such as surgery or radiation therapy. You may also stay on your current treatment if there's some benefit.

A different targeted therapy may be started in place of your current treatment. For cancer that has spread to the brain and is causing symptoms, repotrectinib, taletrectinib, and lorlatinib (Lorbrena) are preferred therapies, while entrectinib after taking crizotinib is sometimes useful. Repotrectinib or taletrectinib is an option for resistant mutations, such as *ROS1* G2032R.

If targeted therapy is not likely to help, your team may recommend the treatments listed in *Chapter 6: Treatment by NSCLC type*.

BRAF V600E mutation

BRAF, a signaling protein, can be overactive, causing cancer cells to grow quickly. A *BRAF* V600E mutation causes its overactivity.

Preferred therapies are combinations of kinase inhibitors.

- Dabrafenib (Tafinlar) with trametinib (Mekinist)
- Encorafenib (Braftovi) with binimetinib (Mektovi).

Dabrafenib and encorafenib stop growth signals from BRAF. MEK is a protein within the same signaling pathway as BRAF. Trametinib and binimetinib stop growth signals from MEK.

If you're not already taking kinase inhibitors, you may stop your current treatment early and start targeted therapy. The other option is to finish your current treatment (including the last phase, called maintenance therapy) and then start targeted therapy.

If dabrafenib plus trametinib makes you too sick, you may receive dabrafenib alone or vemurafenib (Zelboraf). Vemurafenib also stops growth signals from BRAF.

Other recommended therapies are described in *Chapter 6: Treatment by NSCLC type*.

In time, the cancer will worsen. After targeted therapy, you may receive treatment based on the type of NSCLC. If not used before, dabrafenib with trametinib or encorafenib with binimetinib may be started.

NTRK gene fusion

Lung cells have a family of 3 cell surface receptors called TRKA, TRKB, and TRKC. *NTRK* genes contain instructions for making TRK. Excess TRK occurs when *NTRK* joins with another gene. The excess receptors cause fast cell growth.

Preferred therapies are TRK inhibitors.

- Larotrectinib (Vitrakvi)
- Entrectinib (Rozlytrek)
- Repotrectinib (Augtyro)

If you're not taking a TRK inhibitor, you may stop your current treatment early and start targeted therapy.

For some people, it's useful to start with treatment described in *Chapter 6: Treatment by NSCLC type*.

In time, the cancer will worsen. After targeted therapy, you may receive treatment based on the type of NSCLC. If not used before, a TRK inhibitor may be started.

MET exon 14 skipping

Some lung cancers have too much of a cell surface receptor called MET. Too much of this receptor causes fast cell growth. One of the causes of excess MET is a deleted (skipped) part of the *MET* gene called exon 14.

Preferred therapies are MET inhibitors.

- Capmatinib (Tabrecta)
- Tepotinib (Tepmetko)

Crizotinib (Xalkori) is useful in certain cases. It inhibits MET and other kinases.

If you're not already taking a MET inhibitor, you may stop your current treatment early and start targeted therapy. But some people start and continue with treatment described in *Chapter 6: Treatment by NSCLC type*.

In time, the cancer will worsen. After targeted therapy, you may receive treatment based on the type of NSCLC. If not used before, a MET inhibitor may be started.

RET gene fusion

A cell surface receptor kinase called RET can be overactive and cause lung cells to multiply. The overactivity is caused by parts of genes switching places and forming a new gene. This is called a gene fusion.

Preferred therapies are RET inhibitors.

- Pralsetinib (Gavreto)
- Selpercatinib (Retevmo)

If you're not already taking a RET inhibitor, you may stop your current treatment early and start targeted therapy.

In time, the cancer will worsen on targeted therapy. A different RET inhibitor called cabozantinib (Cometriq, Cabometyx) may be started. The other option is to start treatment described in *Chapter 6: Treatment by NSCLC type*.

HER2 mutation

Lung cancer cells have a receptor on their surface called HER2. Certain mutations in the *HER2* gene (also called the *ERBB2* gene) cause the receptor to be overactive. HER2 overactivity makes the cancer cells grow quickly.

Starting treatment with targeted therapy

Lung cancers with *HER2* mutations may be first treated with zongertinib (Hernexeos).

If the cancer grows, the next options include antibody-drug conjugates called fam-trastuzumab deruxtecan-nxki (Enhertu) and ado-trastuzumab emtansine (Kadcyla). A third option is treatment discussed in *Chapter 6: Treatment by NSCLC type*.

Starting treatment based on NSCLC type

A second option is to start treatment based on type of NSCLC as described in Chapter 6.

If the cancer grows, there are several options:

- Preferred therapies are zongertinib, fam-trastuzumab deruxtecan, and sevabertinib (Hyrnuo).
- Ado-trastuzumab emtansine (Kadcyla) is useful in some cases.

Treatment after targeted therapy is again based on the type of NSCLC. Options are second-line therapies discussed in Chapter 6.

NRG1 gene fusion

An *NRG1* gene fusion is a rare driver mutation of NSCLC. It's more common in adenocarcinoma than in other types of NSCLC. The abnormal *NRG1* gene fusion makes a protein that increases cell growth signals from the HER3-HER2 pathway.

Lung cancers with *NRG1* gene fusions are first treated by type of NSCLC. Treatment options are described in *Chapter 6: Treatment by NSCLC type*.

If the cancer grows after the first treatment, you may receive a HER2-HER3 antibody called zenocutuzumab-zbco (Bizengri). Treatment options after zenocutuzumab are the second-line therapies listed in Chapter 6.

Key points

- A driver mutation causes normal cells to become cancer cells.
- Targeted therapy is a newer option for treating lung cancers with driver mutations. Therapies are specific to the type of driver mutation.
- When cancer grows during targeted therapy, treatment may be continued or switched to a different targeted therapy.
- When targeted therapy is not likely to help, you may receive treatment based on the type of NSCLC.

Questions to ask

- Which driver mutations does the lung cancer have?
- How did you decide what treatment is best for me?
- What is the schedule of my treatment?
- What are the common and serious side effects of my treatment?
- Is there a clinical trial that's a good fit for me?

What's next?

It's common to have many concerns about cancer. Your care team will support you. More sources of support are listed in *Chapter 7: Other resources*.

"It is important to understand what you are going through. If your doctor says something you do not understand, don't just nod like you do. Let your doctor know. You can say, 'I don't fully understand what you just said. Can you please explain it in simpler terms?'"

5

Treatment based on abnormal PD-L1 levels

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42 Questions to ask

Some lung cancers avoid death by stopping cancer-fighting T cells. Immunotherapy restores the killing ability of these T cells. Read this chapter to learn about this survival skill of cancer cells and how it can be stopped.

What is PD-L1?

In short, PD-L1 is a protein on the surface of lung cancer cells. It stops the body from killing cancer cells. To better understand treatment for non-small cell lung cancer (NSCLC), let's look in more detail at how PD-L1 works.

T cells kill cancer cells

The immune system is the body's natural defense against disease, including cancer. A type of white blood cell called T cells are a key part of this system. T cells that kill cancer cells are called cytotoxic, or killer, T cells.

PD-1 limits T-cell activity

The immune system has brakes that control an immune response. These brakes are called immune checkpoints. They ensure that the immune response doesn't damage the body. PD-1 is a brake on T cells.

PD-L1 turns on PD-1

In some people with NSCLC, the immune brakes are overused. The PD-1 brake is activated when PD-L1 on cancer cells attaches to it.

Cancer treatment stops PD-L1

Immune checkpoint inhibitors are a type of immunotherapy that keeps the immune brakes turned off. In other words, they enable T cells to kill cancer cells.

Planning treatment

Your care team will decide if immune checkpoint inhibitors are an option for you.

Immune checkpoint inhibitors may not be safe if:

- You have an autoimmune disease, are taking medications that suppress your immune system, or have had an organ transplant. Safer treatments are discussed in *Chapter 6: Treatment by NSCLC type*.
- Your performance status is either 3 or 4 on the Eastern Cooperative Oncology Group scoring system. These scores suggest that the cancer treatments listed in this chapter would likely be too harmful, so your team may recommend supportive care instead.

Immune checkpoint inhibitors may not be the best option for you because they have less benefit for lung cancers with an:

- *EGFR* exon 19 deletion or L858R mutation
- *ALK*, *RET,* or *ROS1* gene fusion

Treatment of these cancers is discussed in *Chapter 4: Treatment of driver mutations*.

If immune checkpoint inhibitors are an option, your team will plan treatment based on PD-L1 level.

- High PD-L1 means that at least half of the cancer cells have PD-L1 (50 percent [50%] or more).
- Low PD-L1 means that less than half of cancer cells have PD-L1 (1% to 49%).

PD-L1 levels are within the normal range when fewer than 1 out of 100 cells have PD-L1 (less than 1%). Treatment of cancers with normal PD-L1 and no driver mutations is discussed in *Chapter 6: Treatment by NSCLC type*.

Types of immune checkpoint inhibitors

There are 3 types of immune checkpoint inhibitors that treat NSCLC.

Two types stop PD-L1 from activating PD-1 and are named after the protein they attach to.

- **PD-1 inhibitors** attach to PD-1 on T cells. PD-1 inhibitors are pembrolizumab (Keytruda), nivolumab (Opdivo), and cemiplimab-rwlc (Libtayo).
- **PD-L1 inhibitors** attach to PD-L1 on cancer cells. PD-L1 inhibitors are atezolizumab (Tecentriq) and durvalumab (Imfinzi).

NSCLC also causes a second type of immune brake, called CTLA-4, to be overused. This brake is overused because lung cancer causes high CTLA-4 levels.

The CTLA-4 brake on T cells is activated when it attaches to B7 on immune cells called dendritic cells.

- **CTLA-4 inhibitors** attach to CTLA-4 and prevent the activation of this brake by B7. CTLA-4 inhibitors are ipilimumab (Yervoy) and tremelimumab-actl (Imjudo).

The immune checkpoint inhibitors listed above are slowly injected into a vein. This slow injection is called an infusion. Some checkpoint inhibitors are also available as injections under the skin.

- Atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza)
- Pembrolizumab and berahyaluronidase alfa-pmph (Keytruda Qlex)

First-line and maintenance therapy

The first regimen you receive is called first-line therapy. A regimen is one or more cancer drugs taken at a specific dose, schedule, and length of time.

If first-line therapy controls cancer growth, you may shift to maintenance therapy. Maintenance therapy includes 1 or 2 drugs from your first-line therapy. Its goal is to stop cancer growth for as long as possible.

NCCN recommendations and preference

In addition to PD-L1 level, NCCN recommendations are based on the type of NSCLC. Some cancer drugs that treat lung adenocarcinoma, large cell carcinoma, and

rare cell types don't treat squamous cell carcinoma.

First-line and maintenance therapies for adenocarcinoma, large cell carcinoma, and rare cell types with high PD-L1 are listed in **Guide 7**; therapies for squamous cell carcinoma with high PD-L1 are in **Guide 8**.

Likewise, options for cancers with low PD-L1 are listed in **Guide 9** and **Guide 10**.

Guide 7
High PD-L1 treatment: Adenocarcinoma, large cell carcinoma, rare cell types

Preferred therapies

- Pembrolizumab followed by maintenance Pembrolizumab
- Carboplatin (or Cisplatin), Pemetrexed plus Pembrolizumab followed by maintenance Pemetrexed plus Pembrolizumab, then Pemetrexed
- Atezolizumab followed by maintenance Atezolizumab
- Cemiplimab-rwlc followed by maintenance Cemiplimab
- Carboplatin (or Cisplatin), Pemetrexed plus Cemiplimab-rwlc followed by maintenance Cemiplimab with or without Pemetrexed

Other recommended therapies

- Carboplatin, Paclitaxel plus Atezolizumab and Bevacizumab followed by maintenance Bevacizumab with or without Atezolizumab
- Albumin-bound Paclitaxel, Carboplatin plus Atezolizumab followed by maintenance Atezolizumab
- Carboplatin (or Cisplatin), Pemetrexed plus Ipilimumab and Nivolumab followed by maintenance Ipilimumab and Nivolumab
- Carboplatin (or Cisplatin), Paclitaxel plus Cemiplimab-rwlc followed by maintenance Cemiplimab
- Albumin-bound Paclitaxel, Carboplatin plus Durvalumab and Tremelimumab-actl followed by maintenance Durvalumab
- Carboplatin (or Cisplatin), Pemetrexed plus Durvalumab and Tremelimumab-actl followed by maintenance Durvalumab with or without Pemetrexed

Therapy used in certain cases

- Ipilimumab plus Nivolumab followed by maintenance Ipilimumab plus Nivolumab

NCCN experts have assigned a level of preference to their recommendations:

- **Preferred therapies** have the most evidence they may work better and be safer than other therapies.
- **Other recommended therapies** can provide effective results but may have less evidence, more side effects, or may not work quite as well as preferred therapies.
- **Therapies used in certain cases** work best for individuals with specific cancer features or health circumstances.

Single immunotherapy

Some lung cancers with PD-L1 are treated with only 1 drug—an immune checkpoint inhibitor.

Pembrolizumab, atezolizumab, and cemiplimab-rwlc are preferred options for all types of NSCLC with high PD-L1.

Guide 8
High PD-L1 treatment: Squamous cell carcinoma

Preferred therapies
• Pembrolizumab followed by maintenance Pembrolizumab
• Albumin-bound Paclitaxel (or Paclitaxel), Carboplatin plus Pembrolizumab followed by maintenance Pembrolizumab
• Atezolizumab followed by maintenance Atezolizumab
• Cemiplimab-rwlc followed by maintenance Cemiplimab
• Carboplatin (or Cisplatin), Paclitaxel plus Cemiplimab-rwlc followed by maintenance Cemiplimab
Other recommended therapies
• Carboplatin, Paclitaxel plus Ipilimumab and Nivolumab followed by maintenance Ipilimumab and Nivolumab
• Albumin-bound Paclitaxel, Carboplatin plus Durvalumab and Tremelimumab-actl followed by maintenance Durvalumab
• Carboplatin (or Cisplatin), Gemcitabine plus Durvalumab and Tremelimumab-actl followed by maintenance Durvalumab
Therapy used in certain cases
• Ipilimumab plus Nivolumab followed by maintenance Ipilimumab and Nivolumab

Pembrolizumab by itself is useful for some cancers with low PD-L1.

Your performance status should be adequate (score of 0, 1, or 2) to receive immune checkpoint inhibitors. Atezolizumab is an exception. It may be an option if your performance status is 3.

Chemoimmunotherapy

Many preferred and other recommended therapies are a pairing of an immune checkpoint inhibitor with platinum-doublet chemotherapy. This combined treatment is called chemoimmunotherapy.

Guide 9
Low PD-L1 treatment: Adenocarcinoma, large cell carcinoma, and rare cell types

Preferred therapies

- Carboplatin (or Cisplatin), Pemetrexed plus Pembrolizumab followed by maintenance Pemetrexed plus Pembrolizumab, then Pemetrexed
- Carboplatin (or Cisplatin), Pemetrexed plus Cemiplimab-rwlc followed by maintenance Cemiplimab with or without Pemetrexed

Other recommended therapies

- Carboplatin, Paclitaxel plus Atezolizumab and Bevacizumab followed by maintenance Bevacizumab with or without Atezolizumab
- Albumin-bound Paclitaxel, Carboplatin plus Atezolizumab followed by maintenance Atezolizumab
- Carboplatin (or Cisplatin), Pemetrexed plus Ipilimumab and Nivolumab followed by maintenance Ipilimumab and Nivolumab
- Ipilimumab plus Nivolumab followed by maintenance Ipilimumab and Nivolumab
- Carboplatin (or Cisplatin), Paclitaxel plus Cemiplimab-rwlc followed by maintenance Cemiplimab
- Albumin-bound Paclitaxel, Carboplatin plus Durvalumab and Tremelimumab-actl followed by maintenance Durvalumab
- Carboplatin (or Cisplatin), Pemetrexed plus Durvalumab and Tremelimumab-actl followed by maintenance Durvalumab with or without Pemetrexed

Therapy used in certain cases

- Pembrolizumab followed by maintenance Pembrolizumab

Chemotherapy kills fast-growing cells including cancer cells. Platinum-doublet chemotherapy consists of 2 types of chemotherapy.

- The platinum-based chemotherapy is either cisplatin or carboplatin.
- The second chemotherapy is one of the following: pemetrexed (Pemrydi RTU, Axtle, Alimta, Pemfexy), paclitaxel, paclitaxel with human albumin (Abraxane), or gemcitabine (Avgemsi).

Platinum-doublet chemotherapy can cause serious side effects, so you must be healthy enough to get this treatment.

Chemoimmunotherapy and bevacizumab

Bevacizumab (Avastin) is a targeted therapy called a VEGF antibody. It stops the growth of blood vessels on tumors. Without blood, cancer cells die.

One of the other recommended therapies for adenocarcinoma, large cell carcinoma, and rare cell types is carboplatin and paclitaxel plus atezolizumab and bevacizumab.

Double immunotherapy

For some lung cancers with high PD-L1, nivolumab plus ipilimumab may be an option.

Guide 10
Low PD-L1 treatment: Squamous cell carcinoma

Preferred therapies

- Albumin-bound Paclitaxel (or Paclitaxel), Carboplatin plus Pembrolizumab followed by maintenance Pembrolizumab
- Carboplatin (or Cisplatin), Paclitaxel plus Cemiplimab-rwlc followed by maintenance Cemiplimab

Other recommended therapies

- Carboplatin, Paclitaxel plus Ipilimumab and Nivolumab followed by maintenance Ipilimumab and Nivolumab
- Ipilimumab plus Nivolumab followed by maintenance Ipilimumab and Nivolumab
- Albumin-bound Paclitaxel, Carboplatin plus Durvalumab and Tremelimumab-actl followed by maintenance Durvalumab
- Carboplatin (or Cisplatin), Gemcitabine plus Durvalumab and Tremelimumab-actl followed by maintenance Durvalumab

Therapy used in certain cases

- Pembrolizumab followed by maintenance Pembrolizumab

Monitoring results

First-line therapies are given in cycles of treatment days, followed by rest days. One cycle typically lasts for several weeks.

NCCN experts recommend checking treatment results after 2 cycles. A CT (computed tomography) scan will be done to check results. Contrast may be used. If the cancer has not worsened, NCCN experts recommend getting CT scans every 2 to 4 cycles.

During maintenance therapy, NCCN experts recommend checking treatment results with a CT scan every 6 to 12 weeks.

Key points

- Some lung cancers have above-normal levels of PD-L1, which stops immune cells from destroying cancer cells.
- Immune checkpoint inhibitors allow immune cells to attack cancer cells.
- Not everyone can take immune checkpoint inhibitors. If you have no restrictions, the one chosen for you will be based on the level of PD-L1 and type of lung cancer.
- If cancer growth slows, you may stay on some drug treatments to increase the time until the cancer grows again. This is called maintenance therapy.

Questions to ask

- Does the lung cancer have normal, low, or high PD-L1 levels?
- How did you decide what treatment is best for me?
- What is the schedule, or cycles, of my treatment?
- What are the common and serious side effects of my treatment?
- Is there a clinical trial that's a good fit for me?

What's next?

Within a few years of first-line therapy, lung cancer starts to grow again in most people. The next treatment is explained in *Chapter 6: Treatment by NSCLC type*.

- If you didn't receive chemoimmunotherapy, read the section in Chapter 6 called "First-line therapy."
- If you had chemoimmunotherapy, read the section in Chapter 6 called "Second-line therapy."

6
Treatment by NSCLC type

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Lung cancer differs among people by the type of affected lung cell. Your treatment will be specific to the type of lung cancer you have. In addition to standard treatment, a clinical trial may be an option for you.

When lung cancer doesn't have a driver mutation or abnormal PD-L1 levels, treatment is mainly based on the type of non-small cell lung cancer (NSCLC). This chapter lists the options for treatment.

Some options in this chapter are also sometimes used to treat NSCLC with driver mutations or abnormal PD-L1 levels as explained in Chapters 4 and 5.

Planning treatment

Your care team will decide what treatments are options for you.

The cancer treatments listed in this chapter may not be safe if your performance status is poor (either 3 or 4 on the Eastern Cooperative Oncology Group scoring system). These scores suggest that cancer treatment would likely be too harmful, so your team may recommend supportive care instead.

If cancer treatment is safe, your team will choose treatment based on your medical history, the type of NSCLC, and biomarkers. Some cancer drugs that treat lung adenocarcinoma, large cell carcinoma, and rare cell types don't treat squamous cell carcinoma.

Types of therapy

Most therapies for NSCLC consist of chemotherapy, immunotherapy, or a combination of both.

Chemotherapy

The classic treatment for widespread metastatic lung cancer is chemotherapy. It kills fast-growing cells, including cancer cells. One or more chemotherapy drugs may be a part of treatment.

Platinum-doublet chemotherapy consists of the following 2 types of chemotherapy:

- The platinum-based chemotherapy is either cisplatin or carboplatin.
- The second chemotherapy is pemetrexed (Axtle, Alimta, Pemfexy, Pemrydi RTU), paclitaxel, paclitaxel with human albumin (Abraxane), etoposide (Etopophos), or gemcitabine (Avgemsi).

Platinum-doublet chemotherapy can cause serious side effects, so you must be healthy enough to get this treatment.

Immune checkpoint inhibitors

Immune checkpoint inhibitors restore the ability of immune cells to kill cancer cells. They treat lung cancer with PD-L1 as explained in Chapter 5. But they may also extend life when lung cancer does not have PD-L1.

There are 3 types of checkpoint inhibitors each named after the protein it attaches to.

- **PD-1 inhibitors** – pembrolizumab (Keytruda), nivolumab (Opdivo), and cemiplimab-rwlc (Libtayo)
- **PD-L1 inhibitors** – atezolizumab (Tecentriq) and durvalumab (Imfinzi)
- **CTLA-4 inhibitors** – ipilimumab (Yervoy) and tremelimumab-actl (Imjudo)

Most drug therapies for metastatic NSCLC are slowly injected into a vein. This slow injection is called an infusion. These checkpoint inhibitors are also available as injections under the skin.

- Atezolizumab and hyaluronidase-tqjs (Tecentriq Hybreza)
- Pembrolizumab and berahyaluronidase alfa-pmph (Keytruda Qlex)

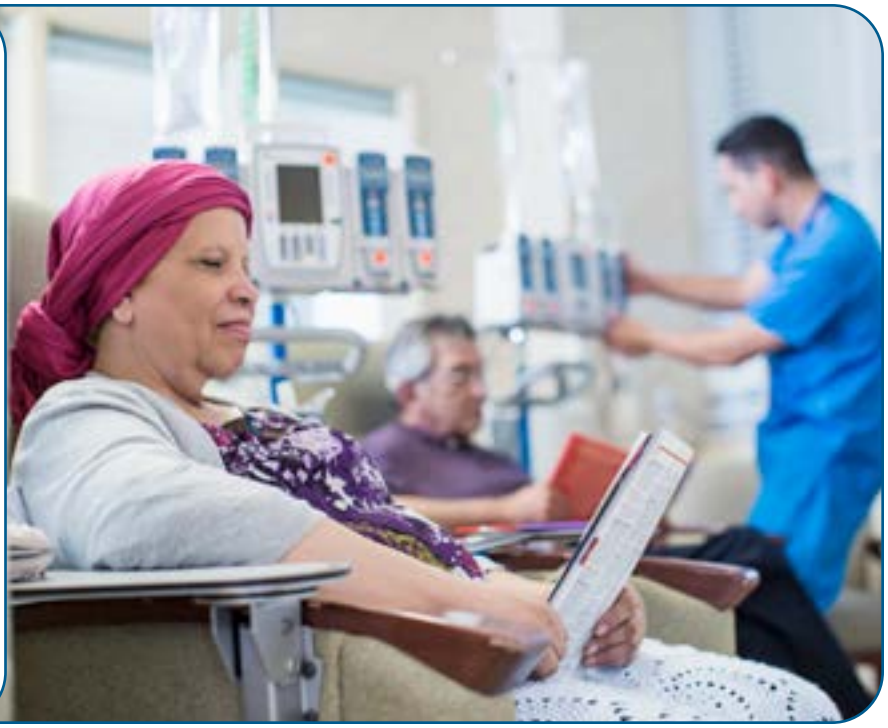
NCCN levels of preference

The next few sections in this chapter list recommended regimens based on NSCLC type. A regimen is one or more cancer drugs that are taken at a specific dose, schedule, and length of time. NCCN experts have assigned a level of preference to their recommendations for therapy:

- **Preferred therapies** have the most evidence they may work better and be safer than other therapies.
- **Other recommended therapies** can provide effective results but may have less evidence, more side effects, or may not work quite as well as preferred therapies.
- **Therapies used in certain cases** work best for individuals with specific cancer features or health circumstances.

Infusions

Drug therapy for lung cancer is often given by injection into a vein. An infusion is a slow drip controlled by a pump that may take hours.



First-line therapy

The first regimen that you will receive is called first-line therapy. First-line therapies for adenocarcinoma, large cell carcinoma, and rare cell types are listed in **Guide 11**; those for squamous cell carcinoma are in **Guide 12**.

Chemoimmunotherapy

Many preferred and other recommended therapies are chemoimmunotherapy regimens.

Chemoimmunotherapy combines immune checkpoint inhibitors with platinum-doublet chemotherapy.

Some regimens for lung adenocarcinoma, large cell carcinoma, and rare cell types also include a targeted therapy called bevacizumab (Avastin). An FDA-approved biosimilar can be taken instead of bevacizumab. A biosimilar is almost an identical drug made by another company.

Guide 11
First-line therapy for metastatic lung adenocarcinoma, large cell carcinoma, and rare cell types

Preferred therapies	• Carboplatin, Pemetrexed plus Pembrolizumab • Cisplatin, Pemetrexed plus Pembrolizumab • Carboplatin (or Cisplatin), Pemetrexed plus Cemiplimab-rwlc
Other recommended therapies	• Carboplatin, Paclitaxel plus Atezolizumab and Bevacizumab • Albumin-bound Paclitaxel, Carboplatin plus Atezolizumab • Ipilimumab plus Nivolumab • Carboplatin (or Cisplatin), Pemetrexed plus Ipilimumab and Nivolumab • Carboplatin (or Cisplatin), Paclitaxel plus Cemiplimab-rwlc • Albumin-bound Paclitaxel, Carboplatin plus Durvalumab and Tremelimumab-actl • Carboplatin (or Cisplatin), Pemetrexed plus Durvalumab and Tremelimumab-actl
Therapies used in certain cases	• Carboplatin, Paclitaxel plus Bevacizumab • Carboplatin, Pemetrexed plus Bevacizumab • Cisplatin, Pemetrexed plus Bevacizumab • Carboplatin and Albumin-bound Paclitaxel, Docetaxel, Etoposide, Gemcitabine, Paclitaxel, or Pemetrexed • Cisplatin and Docetaxel, Etoposide, Gemcitabine, Paclitaxel, or Pemetrexed • Gemcitabine and either Docetaxel or Vinorelbine • Single-drug chemotherapy

Immune checkpoint inhibitors

If you can't have platinum-doublet chemotherapy, you may receive only immune checkpoint inhibitors. Other recommended therapies include nivolumab plus ipilimumab.

Your performance status should be adequate (score of 0, 1, or 2) to receive immune checkpoint inhibitors. Atezolizumab is an exception. It may be an option if your performance status is 3.

Chemotherapy

When immune checkpoint inhibitors are not an option, chemotherapy is used for treatment.

Platinum-doublet chemotherapy is used most often. Bevacizumab is a part of some platinum-doublet regimens.

Additional options include gemcitabine with either docetaxel or vinorelbine. And there are several options for single-drug chemotherapy.

- Albumin-bound paclitaxel
- Docetaxel
- Gemcitabine
- Paclitaxel
- Pemetrexed (only for non-squamous cell types)

Guide 12
First-line therapy for metastatic squamous cell carcinoma of the lung

Preferred therapies	• Carboplatin, Paclitaxel plus Pembrolizumab • Albumin-bound Paclitaxel, Carboplatin plus Pembrolizumab • Carboplatin (or Cisplatin), Paclitaxel plus Cemiplimab-rwlc
Other recommended therapies	• Ipilimumab plus Nivolumab • Carboplatin, Paclitaxel plus Ipilimumab and Nivolumab • Albumin-bound paclitaxel, Carboplatin plus Durvalumab and Tremelimumab-actl • Carboplatin (or Cisplatin), Gemcitabine plus Durvalumab and Tremelimumab-actl
Therapies used in certain cases	• Carboplatin and Albumin-bound Paclitaxel, Docetaxel, Gemcitabine, Paclitaxel, or Etoposide • Cisplatin and Docetaxel, Etoposide, Gemcitabine, or Paclitaxel • Gemcitabine and either Docetaxel or Vinorelbine • Single-drug chemotherapy

Monitoring results

First-line therapy is given in cycles of treatment days, followed by rest days. One cycle typically lasts for 3 to 4 weeks.

In general, first-line therapy is given for 4 cycles. If treatment isn't making you too sick, a total of 6 cycles may be completed.

After 2 cycles, your team will assess the results. The extent of the cancer can be seen on CT (computed tomography) scans. Contrast may be used. The CT scan will be repeated after another 2 to 4 cycles.

Maintenance therapy

If first-line therapy controls cancer growth, you may receive maintenance therapy after that regimen is finished. The goal of maintenance therapy is to stop the growth of cancer for as long as possible.

Guide 13
Maintenance therapy for metastatic NSCLC

Adenocarcinoma, large cell carcinoma, and rare cell types	Continuation maintenance • Bevacizumab • Pemetrexed • Pemetrexed plus Bevacizumab • Pemetrexed plus Pembrolizumab • Atezolizumab plus Bevacizumab • Ipilimumab plus Nivolumab • Atezolizumab • Gemcitabine • Cemiplimab-rwlc • Durvalumab • Pemetrexed plus Cemiplimab-rwlc or Durvalumab Switch maintenance • Pemetrexed
Squamous cell lung cancer	Continuation maintenance • Pembrolizumab • Nivolumab plus Ipilimumab • Gemcitabine • Cemiplimab-rwlc • Durvalumab

You may stay on at least one of the medicines. This is called continuation maintenance.

Another option is changing to a medicine that you didn't take as a first-line therapy. This is called switch maintenance.

Options for maintenance therapy are discussed in **Guide 13**.

You may stay on maintenance therapy for 2 years if first-line therapy includes immune checkpoint inhibitors.

Second-line therapy

In time, lung cancer often starts to grow again after first-line therapy. Options are listed in **Guide 14**.

Immune checkpoint inhibitors

Immune checkpoint inhibitors are a preferred therapy if you haven't received this treatment before. If the cancer grew while you were taking a checkpoint inhibitor, switching to another checkpoint inhibitor is not advised.

Chemotherapy

Other recommended therapies include chemotherapy. Options are docetaxel plus a targeted therapy called ramucirumab (Cyramza) and single-drug chemotherapy.

Guide 14
Second-line therapy for metastatic NSCLC

Preferred therapy when you haven't had an immune checkpoint inhibitor before	• Nivolumab • Pembrolizumab • Atezolizumab
Other recommended therapies whether you've had an immune checkpoint inhibitor or not	• Docetaxel • Gemcitabine • Docetaxel plus Ramucirumab • Albumin-bound Paclitaxel • Fam-trastuzumab deruxtecan-nxki for high HER2 levels • Pemetrexed only for lung adenocarcinoma, large cell carcinoma, and rare cell types • Telisotuzumab vedotin-tllv only for lung adenocarcinoma, large cell carcinoma, and rare cell types with high HGFR levels

Antibody-drug conjugates

An antibody-drug conjugate binds to certain cancer cells and then releases a drug that attacks the cancer.

Fam-trastuzumab deruxtecan-nxki (Enhertu) is an antibody-drug conjugate for NSCLC with high levels (overexpression) of HER2.

Telisotuzumab vedotin-tllv (Emrelis) is an option for adenocarcinomas, large cell, and rare types of lung cancer with high levels (overexpression) of HGF receptor (c-Met) and no *EGFR* driver mutations.

Monitoring and maintenance

Your team will monitor your treatment results. NCCN experts recommend a CT scan every 6 to 12 weeks. Contrast may be used.

If an immune checkpoint inhibitor is part of second-line therapy, you can stay on maintenance therapy as long as the cancer isn't growing.

Clinical trials

Another way to obtain treatment is through a clinical trial.

A clinical trial is a type of medical research study. After being developed and tested in a lab, potential new ways of fighting cancer need to be studied in people.

If found to be safe and effective in a clinical trial, a drug, device, or treatment approach may be approved by the U.S. FDA.

Everyone with cancer should carefully consider all of the treatment options available for their cancer type, including standard treatments and clinical trials. Talk to your doctor about whether a clinical trial may make sense for you.

Phases

Most cancer clinical trials focus on treatment and are done in phases.

- **Phase 1** trials study the safety and side effects of an investigational drug or treatment approach.
- **Phase 2** trials study how well the drug or approach works against a specific type of cancer.
- **Phase 3** trials test the drug or approach against a standard treatment. If the results are good, it may be approved by the FDA.
- **Phase 4** trials study the safety and benefit of an FDA-approved treatment.

Who can enroll?

It depends on the clinical trial's rules, called eligibility criteria. The rules may be about age, cancer type and stage, treatment history, or general health. They ensure that participants are alike in specific ways and that the trial is as safe as possible for the participants.

Informed consent

Clinical trials are managed by a research team. This group of experts will review the study with you in detail, including its purpose and the risks and benefits of joining. All of this information is also provided in an informed consent form. Read the form carefully and ask questions before signing it. Take time to discuss it with people you trust. Keep in mind that you can leave and seek treatment outside of the clinical trial at any time.

Will I get a placebo?

Placebos (inactive versions of real medicines) are almost never used alone in cancer clinical trials. It's common to receive either a placebo with a standard treatment, or a new drug with a standard treatment. You will be informed, verbally and in writing, if a placebo is part of a clinical trial before you enroll.

Are clinical trials free?

There is no fee to enroll in a clinical trial. The study sponsor pays for research-related costs, including the study drug. But you may need to pay for other services, like transportation or childcare, due to extra appointments. During the trial, you will continue to receive standard cancer care. This care is often covered by insurance.

Finding a clinical trial

In the United States

NCCN Cancer Centers
NCCN.org/cancercenters

The National Cancer Institute (NCI)
cancer.gov/about-cancer/treatment/clinical-trials/search

Worldwide

The U.S. National Library of Medicine (NLM)
clinicaltrials.gov

Need help finding a clinical trial?

NCI's Cancer Information Service (CIS)
1.800.4.CANCER (1.800.422.6237)
cancer.gov/contact

Key points

- Treatment of lung cancer without treatable biomarkers is based on the type of NSCLC.
- Whole-body drug therapy is used to treat people who are fairly healthy. If therapy may be harmful, supportive care is generally recommended.
- If safe and effective, chemoimmunotherapy or immunotherapy is recommended for the first treatment of lung cancer based on the type of NSCLC. Another option is a chemotherapy regimen.
- You may receive between 4 and 6 cycles of treatment. Your team will monitor the results of your treatment.
- Maintenance therapy slows cancer growth. It consists of one or more drugs from your first treatment.
- The next treatment options for lung cancer include an immune checkpoint inhibitor or a chemotherapy regimen. Cancers with high HER2 or HGFR levels may be treated with an antibody-drug conjugate.
- New ways of fighting cancer are studied among people in clinical trials. A clinical trial may be an option in addition to standard treatment.

Questions to ask

- How did you decide what treatment is best for me?
- What are the common and serious side effects of my treatment?
- What is the schedule, or cycles, of my treatment?
- When will the results of my treatment be assessed?
- Is there a clinical trial that's a good fit for me?

What's next?

It's common to have many concerns about cancer. Your care team will support you. More sources of support are listed in *Chapter 7: Other resources*.

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Other resources

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54 What else to do

54 Where to get help

55 Questions to ask

Want to learn more? Here's how you can get additional help.

What else to know

This guide helps you know your options so you can make informed decisions and improve your cancer care. But it's not the only resource that you have.

Ask for as much information and help as you need. Many people are interested in learning more about:

- Support groups in the community and online
- Second opinions
- Clinical trials
- Getting financial help
- Making an advance care plan

What else to do

Your health care center can help you with next steps. It often has on-site resources to help meet your needs and find answers to your questions. Health care centers can also inform you of resources in your community.

In addition to help from your providers, the resources listed in the next section provide support for many people like yourself.

Where to get help

Look through the list below and visit the provided websites to learn more about these organizations.

A Breath of Hope Lung Cancer Foundation
abreathofhope.org

American Lung Association
lung.org

Cancer*Care*
cancercare.org

Cancer Hope Network
cancerhopenetwork.org

Cancer Survivor Care
cancersurvivorcare.org

Caring Ambassadors Program
lungcancercap.org

Free Me from Lung Cancer
freemefromlungcancer.org

Go2 Foundation for Lung Cancer
go2foundation.org

GRACE
cancergrace.org

Imerman Angels
imermanangels.org

LiveLung (Dusty Joy Foundation)
dustyjoy.org

Lung Cancer Action Network (LungCAN)
lungcan.org

Lung Cancer Foundation of America
lcfamerica.org

Lung Cancer Research Foundation
lungcancerresearchfoundation.org

LUNGevity
lungevity.org

PAN Foundation
panfoundation.org

Triage Cancer
triagecancer.org

Questions to ask

- Who can I talk to about help with housing, food, and other basic needs?
- What assistance is available for transportation, childcare, and home care?
- Who can tell me what my options for health insurance are and assist me with applying for insurance coverage?
- How much will I have to pay for my treatment and what help is available for these costs?
- How can I connect with others and build a support system?

Take our survey and help make the NCCN Guidelines for Patients better for everyone!

NCCN.org/patients/comments

Words to know

adenocarcinoma
A cancer of cells that line organs and make fluids or hormones.

adrenal gland
A small organ on top of each kidney that makes hormones.

antibody-drug conjugate
A drug that attaches to certain cells and releases a drug to kill the cells.

antibody therapy
A drug that stops signals from proteins on the surface of cells.

biomarker
Any molecule in your body that can be measured to assess your health.

biopsy
A procedure that removes fluid or tissue samples to be tested for a disease.

cancer stage
A rating of the outlook of a cancer based on its growth and spread.

carcinoma
A cancer of cells that line the inner or outer surfaces of the body.

chemistry profile
A lab test of the amount of 8 chemicals in a sample of blood. Also called metabolic panel.

chemoradiation
A cancer treatment with both cell-killing drugs and high-energy rays.

chemotherapy
Treatment with cancer drugs that kill fast-growing cells.

clinical trial
A type of research that assesses how well health tests or treatments work in people.

complete blood count (CBC)
A lab test that measures the parts of the blood.

computed tomography (CT)
An imaging scan that uses x-rays from many angles to make cross-sectional pictures of the inside of the body.

continuation maintenance
A treatment phase using one or more first-line drugs to prolong good treatment results.

contrast
A substance put into the body to make clearer pictures during imaging.

diagnosis
An identification of an illness based on tests.

gene
Coded instructions in cells for making new cells and controlling how cells behave.

immune checkpoint inhibitor
A drug that restores the immune system's ability to destroy cancer cells.

immunohistochemistry
A special lab test done on a tissue sample.

kinase inhibitor
A drug that stops cell proteins involved in cell growth so new cells aren't made.

large cell carcinoma
A cancer of lung cells that are larger than other cells in the airways.

lymph node
A small, disease-fighting structure that is located throughout the body.

magnetic resonance imaging (MRI)
An imaging scan that uses radio waves and powerful magnets to make pictures of the inside of the body.

maintenance therapy
A treatment phase that is given to prolong good treatment results.

medical history
A report of all your health events and medications.

metastasis
The spread of cancer from the first tumor to a new site.

mutation
Abnormal changes in coded instructions within cells (genes).

non-small cell lung cancer (NSCLC)
A cancer that starts in lung cells that are not small.

pathologist
A doctor who's an expert in testing cells and tissue to find disease.

performance status
A rating of one's ability to do daily activities.

pericardiocentesis
A procedure that removes fluid from around the heart with a needle.

physical exam
A review of the body by a health expert for signs of disease.

platinum-doublet chemotherapy
A treatment with two cell-killing drugs, one of which contains the chemical platinum.

positron emission tomography (PET)
A test that uses radioactive material to see the shape and function of body parts.

positron emission tomography/computed tomography (PET/CT)
A test that uses 2 picture-making methods to show the shape and function of body parts.

pulmonary function tests
A set of breathing tests to test the strength of the lungs.

radiation oncologist
A doctor who's an expert in treating cancer with radiation.

radiation therapy
A treatment that uses intense energy to kill cancer cells.

side effect
An unhealthy or unpleasant physical or emotional response to treatment.

small cell lung cancer (SCLC)
A cancer of small, round lung cells.

squamous cell carcinoma
A type of cancer of thin and flat cells that line the surface of organs.

supportive care
Health care that includes symptom relief but not cancer treatment. Also called palliative care.

surgery
An operation to remove or repair a part of the body.

switch maintenance
A treatment phase with a new drug that is given to prolong good treatment results.

targeted therapy
A drug treatment that impedes the growth process that's specific to cancer cells.

thoracic radiologist
A doctor who's an expert in reading imaging tests of the chest.

thoracoscopy
A procedure to do work in the chest with a device passed through a small cut in the skin. Also called video-assisted thoracoscopic surgery (VATS).

transthoracic needle aspiration (TTNA)
A procedure that removes tissue samples with a thin needle guided through the ribs.

We want your feedback!

Our goal is to provide helpful and easy-to-understand information on cancer. Take our survey to let us know what we got right and what we could do better.
NCCN.org/patients/feedback

NCCN Contributors

This patient guide is based on the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Non-Small Cell Lung Cancer, Version 5.2026. It was adapted, reviewed, and published with help from the following people:

Dorothy A. Shead, MS
Senior Director
Patient Information Operations

Laura J. Hanisch, PsyD
Patient Information Program Manager

Lisa Diehl
Production Layout Artist

The NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Non-Small Cell Lung Cancer, Version 5.2026 were developed by the following NCCN Panel Members:

Gregory J. Riely, MD, PhD/Chair
Memorial Sloan Kettering Cancer Center

Douglas E. Wood, MD/Vice Chair
Fred Hutchinson Cancer Center

Dara L. Aisner, MD, PhD
University of Colorado Cancer Center

Andrea L. Axtell, MD, MPH
University of Wisconsin
Carbone Cancer Center

Jessica R. Bauman, MD
Fox Chase Cancer Center

Ankit Bharat, MD
Robert H. Lurie Comprehensive Cancer
Center of Northwestern University

Joe Y. Chang, MD, PhD
The University of Texas
MD Anderson Cancer Center

Aakash Desai, MD, MPH
O'Neal Comprehensive

Cancer Center at UAB
Thomas J. Dilling, MD, MS

Moffitt Cancer Center
Jonathan Dowell, MD

UT Southwestern Simmons
Comprehensive Cancer Center

Gregory A. Durm, MD, MS
Indiana University Melvin and Bren Simon
Comprehensive Cancer Center

Scott Gettinger, MD
Yale Cancer Center/Smilow Cancer Hospital

Travis E. Grotz, MD
Mayo Clinic Comprehensive Cancer Center

Matthew A. Gubens, MD, MS
UCSF Helen Diller Family
Comprehensive Cancer Center

Aditya Juloori, MD
The UChicago Medicine
Comprehensive Cancer Center

Rudy P. Lackner, MD
Fred & Pamela Buffett Cancer Center

Michael Lanuti, MD
Mass General Cancer Center

Benjamin Levy, MD
Johns Hopkins Kimmel Cancer Center

Jules Lin, MD
University of Michigan Rogel Cancer Center

Billy W. Loo, Jr, MD, PhD
Stanford Cancer Institute

Christine M. Lovly, MD, PhD
Vanderbilt-Ingram Cancer Center

Fabien Maldonado, MD
Vanderbilt-Ingram Cancer Center

Daniel Morgensztern, MD
Siteman Cancer Center at Barnes-
Jewish Hospital and Washington
University School of Medicine

Trey C. Mullikin, MD
Duke Cancer Institute

Thomas Ng, MD
The University of Tennessee
Health Science Center

*Dawn Owen, MD, PhD
Mayo Clinic Comprehensive Cancer Center

Dwight H. Owen, MD, MSc
The Ohio State University Comprehensive
Cancer Center - James Cancer Hospital
and Solove Research Institute

Tejas Patil, MD
University of Colorado Cancer Center

Patricio M. Polanco, MD
UT Southwestern Simmons
Comprehensive Cancer Center

Jonathan Riess, MD, MS
UC Davis
Comprehensive Cancer Center

Ana Lucia Ruano Mendez
Huntsman Cancer Institute
at the University of Utah

*Theresa A. Shapiro, MD, PhD
Johns Hopkins Kimmel Cancer Center

Aditi P. Singh, MD
Abramson Cancer Center
at the University of Pennsylvania

James Stevenson, MD
Case Comprehensive Cancer Center/
University Hospitals Seidman Cancer Center
and Cleveland Clinic Taussig Cancer Institute

*Alda Tam, MD
The University of Texas
MD Anderson Cancer Center

Tawee Tanvetyanon, MD, MPH
Moffitt Cancer Center

Jane Yanagawa, MD
UCLA Jonsson
Comprehensive Cancer Center

Edwin Yau, MD, PhD
Roswell Park Comprehensive Cancer Center

Karen Yun, MD
UC San Diego Moores Cancer Center

NCCN

Kristina Gregory, RN, MSN
Senior Vice President
Clinical Information Programs

Lisa Hang, PhD
Oncology Scientist/Senior Medical Writer

* Reviewed this patient guide. For disclosures, visit NCCN.org/disclosures.

NCCN Cancer Centers

For contact information, visit NCCN.org/cancercenters.

Abramson Cancer Center
at the University of Pennsylvania
Philadelphia, Pennsylvania

Case Comprehensive Cancer Center/
University Hospitals Seidman Cancer Center and
Cleveland Clinic Taussig Cancer Institute
Cleveland, Ohio

City of Hope National Medical Center
Duarte, California

Dana-Farber/Brigham and Women's Cancer Center |
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Boston, Massachusetts

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Jacksonville, Florida
Rochester, Minnesota

Memorial Sloan Kettering Cancer Center
New York, New York

Moffitt Cancer Center
Tampa, Florida

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Birmingham, Alabama

Robert H. Lurie Comprehensive Cancer Center
of Northwestern University
Chicago, Illinois

Roswell Park Comprehensive Cancer Center
Buffalo, New York

Siteman Cancer Center at Barnes-Jewish Hospital
and Washington University School of Medicine
St. Louis, Missouri

St. Jude Children's Research Hospital/
The University of Tennessee Health Science Center
Memphis, Tennessee

Stanford Cancer Institute
Stanford, California

The Ohio State University Comprehensive Cancer Center -
James Cancer Hospital and Solove Research Institute
Columbus, Ohio

The UChicago Medicine Comprehensive Cancer Center
Chicago, Illinois

The University of Texas MD Anderson Cancer Center
Houston, Texas

UC Davis Comprehensive Cancer Center
Sacramento, California

UC San Diego Moores Cancer Center
La Jolla, California

UCLA Jonsson Comprehensive Cancer Center
Los Angeles, California

UCSF Helen Diller Family Comprehensive Cancer Center
San Francisco, California

University of Colorado Cancer Center
Aurora, Colorado

University of Michigan Rogel Cancer Center
Ann Arbor, Michigan

University of Wisconsin Carbone Cancer Center
Madison, Wisconsin

UT Southwestern Simmons
Comprehensive Cancer Center
Dallas, Texas

Vanderbilt-Ingram Cancer Center
Nashville, Tennessee

Yale Cancer Center/Smilow Cancer Hospital
New Haven, Connecticut

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3025 Chemical Road, Suite 100
Plymouth Meeting, PA 19462
215.690.0300

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